



Artificial Intelligence and Machine Learning

Module 1: Overview of AI and ML

Video 1: Module Overview

Today, we stand on the brink of our evolution, not with the technology itself, but how we harness data to transform our businesses. The question I pose to you now is, are you ready to redefine the boundaries of your market with data intelligence? Welcome everyone. Today marks an exciting opportunity for us all to delve into a field that is not just transforming our digital landscapes, but also reshaping how we think about technology and its capabilities. This is your introduction to artificial intelligence and machine learning. Technologies are pivotal in the era of digital innovation.

Our session today is structured to not only introduce you to AI and machine learning, but to deepen your understanding to its impact and practical applications across various industries. Here's what we'll cover. We'll start at the foundation, breaking down the jargon and complexity into clear, understandable concepts. What is AI? How do you differ from machine learning? And importantly, what are the types of AI systems that you might encounter in your daily work or life? Once we've grasp the basics will transition into exploring how these technologies are being applied today. From healthcare to finance, from retail to manufacturing, AI and machine learning are not just buzzwords but tools that are driving real and measurable change. Imagine a world where doctors can predict diseases before any symptoms appear, or where you are shopping experience is completely personalised to your taste and budget, all thanks to AI. Today, we are going to show you how these are not just possibilities, but realities being implemented by businesses and organisations around the world.

This agenda is designed not to just inform you but to inspire you. As we walk through these topics, think about emerging trends in AI and machine learning might influence your career, enhance your business, or even change your approach to new challenges. Are you ready to embark in this journey with me? Let's dive in and discover the incredible world of AI and machine learning together. Imagine a world where every decision, from the mundane to the monumental is enhanced by intelligence far beyond our own. This is not the realm of science fiction. It's the promise of AI and machine learning here and now. Hence, the best way to predict the future is to create it now. This isn't just a motivational code, it embodies the essence of what we're going to dive into today, AI and machine learning or ML. As we venture into journey together, we will explore various aspects of machine learning. We look at the what, the why, and the how with a focus on practical, real world applications. And yes, there will be an interactive session where you will get a hands on experience with different AI and ML types. I invite you all to join me in this exploration to demystify the



complex and uncover the potential of AI and machine learning. Let's create the future together. Thank you and let's dive in.

Welcome to a glimpse into the Future, the emerging tech and impact radar for 2024. Today, we will dive into not just what the future holds, but how it's already reshaping industries globally. We dive into the core of what is driving today's most innovative business solutions, artificial intelligence and machine learning. But we won't just skim the surface, we'll explore these fundamentals through a lens that is directly impacts our businesses. Let's begin by deciphering the essence of data science, the science of extracting knowledge and actionable insights not obvious from data to define further, data science is composed of domain knowledge, business acumen, and experience, which are valuable to the business. Computer science, which involves data engineering and wrangling, coding or even ethical hacking, etcetera. Lastly, math and statistics, which are algorithms and numerical techniques to derive insights. Now imagine each pairing of these fields, business and computer science combined to create traditional software programs and applications tailored for specific tasks. When business meets math and statistics, we venture into the realm of traditional research, rigorous analysis and understanding of domain specific problems. And then there's the union of computer science and math and statistics. Here lies the foundation of machine learning, where algorithm learns from and make predictions based on data. What is the sweet spot where they all meet? That's data science. But data science transcends these intersections. It's an ultimate that turns raw data into information, information into insights, and insights into tangible outcomes. It's about leveraging all three areas to not only answer the what, but to explore the why and predict the what's next. This is the blueprint of data science, a fusion of expertise that empowers us to drive change and shape features.

Video 2: Fundamentals of Data Science

Now, let's talk about data science as an art. Yes, you heard it right: artistic. Data science isn't just about crunching numbers. It's about telling a story, painting a picture with data. It's where creativity meets analytics. Imagine a painter, but instead of a brush, they wield algorithms. Every stroke is a line of code, every colour a piece of data. It's not just science, it's a masterpiece in the making. We, as data scientists, are artists who reveal dancing, who tell stories hidden from the numbers, who compose the symphony of insights. But an artist must also understand their audience. In data science, we must ensure our creations, our models and insights, are not only accurate but understandable and actionable. They must resonate with the decision-makers, with the stakeholders, with the very people whose lives and work they are meant to enhance. Now, let's paint a broader picture. Imagine data science as the canvas and what are the brush strokes? They are AI, ML and DL. Each of them adds depth and detail to our masterpiece. Imagine you're steering a company that predicts market trends. Artificial intelligence is your autopilot, interpreting complex data to

make human-like decisions. It navigates through the noise, seeking patterns much like a seasoned trader.

Machine learning, the engine that powers your autopilot. It's your team of analysts learning from the past market movements to predict the future trends without needing explicit commands on how to do so. Deep learning, that's even deeper. It's the neural network, the brain of operations, scrutinising layers upon layers of data, uncovering insights that are even the most experienced analysts might miss. In the grand scheme of things, data science is the universe in which this stars align. It encompasses all the tools, techniques, and the knowledge to turn vast amounts of data into actionable insights. So, we've got our AI, the dream, ML, the magic that brings dream to life, and DL, the deep magic, the enigmatic core. It's a bit of like a Matryoshka Dal, with each layer revealing another more intricate one. And in the heart, that's deep learning, the essence of our modern AI marvels. Now that we've set the stage, let's dive deeper into ML, the core of our discussion, and see how we that's just transforming our data, but our world. Ready to dive in? Every day, we interact with artificial intelligence in a way that are subtle yet incredibly impactful. Today, I want you to focus on two examples that most of us use even without thinking about sophisticated AI working in the background, Waze for navigation, and Gmail for email management. Let's start with ways. How does it manage to outsmart traffic consistently, saving us precious minutes or even hours?

The answer lies in the real-time use of AI. We collect massive amounts of data from users in real-time. It uses AI algorithms to analyse this data, predict traffic conditions, and then recommend the fastest route to the users. What's remarkable here isn't just the data collection, but how AI turns this data into actionable advice instantaneously. Now, let's switch gears to Gmail. Have you noticed how it offers suggestions to finish our sentences as you typed? Or how it proposed quick responses to emails? This is AI at work through Gmail's Smart Compose and Smart reply features. Netflix, a leader in streaming entertainment, utilises machine learning to personalise viewing experiences for millions of users globally. But how does machine learning actually function in this setting? Netflix employs sophisticated algorithms to analyse your watching habits, what you watch, when you watch, and how often you watch. Shows or movies. This data is used to predict and recommend content that aligns closely with your preferences, ensuring that you find your next favourite show without needing to search for it. Switching gears to Facebook, a platform where billions of images are uploaded daily, Facebook's machine learning technology powers an advanced face recognition system. This system can identify and tag users in photos with astonishing accuracy. By analysing patterns and features in the faces, such as the distance between the eyes, the shape of cheekbones, and other facial landmarks, Facebook creates a unique facial fingerprint. This technology not only enhances our interactions by automating photo tagging but also improves security by helping detect fraudulent use of images. These examples illustrate not just the capabilities of machine learning but its integration into our daily lives, enhancing convenience and

security. Netflix recommendations save us time by directing us to content we are likely to enjoy, increasing our satisfaction and retention on the platform simultaneously. Now, let's switch gears to Gmail. Have you noticed how it offers suggestions to finish our sentences as you typed? Or how it proposed quick responses to emails? This is AI at work through Gmail's Smart Compose and Smart reply features. Netflix, a leader in streaming entertainment, utilises machine learning to personalise viewing experiences for millions of users globally. But how does machine learning actually function in this setting? Netflix employs sophisticated algorithms to analyse your watching habits, what you watch, when you watch, and how often you watch. Shows are movies. This data is used to predict and recommend content that aligns closely with your preferences, ensuring that you find your next favourite show without needing to search for it. Switching gears to Facebook, a platform where billions of images are uploaded daily, Facebook's machine learning technology powers an advanced face recognition system. This system can identify and tag users in photos with astonishing accuracy. By analysing patterns and features in the faces, such as the distance between the eyes, the shape of cheekbones, and other facial landmarks, Facebook creates a unique facial fingerprint. This technology not only enhances our interactions by automating photo tagging but also improves security by helping detect fraudulent use of images. These examples illustrate not just the capabilities of machine learning but its integration into our daily lives, enhancing convenience and security. Netflix recommendations save us time by directing us to content we are likely to enjoy, increasing our satisfaction and retention on the platform simultaneously. Facebook's face recognition supports both social connectivity and platform safety. Imagine the possibilities as these technologies continue to evolve. The question for us is not if but how we leverage machine learning to transform our services and products. What aspects of your daily operations could be revolutionised by similar ML applications? Deep learning models are inspired by our brains architecture, specifically the way neurons interact. These models use layers of algorithms, each designed to recognise patterns that humans might miss. Think of these layers as simplified mathematical model of how neurons process information. Take Amazon Alexa, a voice-activated assistant that uses deep learning to understand and respond to your commands. Every time you ask Alexa a question, deep learning algorithms process your speech, break it down into understandable segments, and deliver our spans that meets your needs. This seamless interaction is powered by training thousands of hours of audio data to interpret human speech accurately. Now, let's shift gears to Tesla's driving car. Tesla's Autopilot system uses deep learning to visually interpret the environment. This system processes data from cameras and sensors to make decisions in real-time, from recognising a stop sign to detecting a pedestrian to navigating curves on the road. It's a prime example of how deep learning takes input from the real world to make complex decisions that can improve safety and efficiency. These examples, Alexa and Tesla, showcase deep learnings ability to enhance and automate our daily lives. Facebook's face recognition supports both social connectivity and platform safety. Imagine the possibilities as these

technologies continue to evolve. The question for us is not if, but how we leverage machine learning to transform our services and products. What aspects of your daily operations could be revolutionised by similar ML applications? Deep learning models are inspired by our brain's architecture, specifically the way neurons interact. These models use layers of algorithms, each designed to recognise patterns that humans might miss. Think of these layers as simplified mathematical model of how neurons process information. Take Amazon Alexa, a voice activated assistant that uses deep learning to understand and respond to your commands. Every time you ask Alexa a question, deep learning algorithms process your speech, break it down into understandable segments, and deliver our spans that meets your needs. This seamless interaction is powered by training thousands of hours of audio data to interpret human speech accurately. Now, let's shift gears to Tesla's driving car. Tesla's Autopilot system uses deep learning to visually interpret the environment. This system processes data from cameras and sensors to make decisions in real-time, from recognising a stop sign to detecting a pedestrian to navigating curves on the road. It's a prime example of how deep learning takes input from the real world to make complex decisions that can improve safety and efficiency. These examples, Alexa and Tesla, showcase deep learnings ability to enhance and automate our daily lives. But the potential extends far beyond, from healthcare, where deep learning diagnosis diseases with precision, to finance where it predicts market trends. The applications are as boundless as our ability to imagine execute them. Imagine having the power to predict the future. Not just any prediction, but one that accounts for 150 trillion different outcomes. Today, we will dive into the realm of forecasting and dynamic pricing, a realm where data not only predicts the future, but shapes it. Dynamic pricing isn't new, but the scale and the sophistication we achieved today are unprecedented. Businesses from airlines to e-commerce use dynamic pricing to adjust the cost of goods and services in real time based on variables such as demand, weather and competition, and even consumer behaviour. Let's take the example of a global e-commerce giant. Think of a company that manages millions of products for Black Friday. The leverage and advanced machine learning model to predict pricing trends and optimise sales. This model process 150 trillion combination possibilities to set the prices that maximise profit while ensuring customer satisfaction. Today, we explore 2 revolutionary AI applications in Fintech, algorithmic trading and fraud detection. Each of these technologies is reshaping how the finance sector operates, enhancing both efficiency and security. Let's start with algorithmic trading. Imagine 3D strategies that execute thousands of orders. A second strategies that can analyse and react to market data faster than any human. This is the reality of algorithmic trading, where AI algorithms use mathematical models to make lightning fast decisions, maximising profit and minimising risks. But the potential extends far beyond, from healthcare, where deep learning diagnosis diseases with precision, to finance where it predicts market trends. The applications are as boundless as our ability to imagine execute them. Imagine having the power to predict the future. Not just any prediction, but one that accounts for 150 trillion different outcomes. Today,

we will dive into the realm of forecasting and dynamic pricing, a realm where data not only predicts the future, but shapes it. Dynamic pricing isn't new, but the scale and the sophistication we achieved today are unprecedented. Businesses from airlines to e-commerce use dynamic pricing to adjust the cost of goods and services in real time based on variables such as demand, weather and competition, and even consumer behavior. Let's take the example of a global e-commerce giant. Think of a company that manages millions of products for Black Friday. The leverage and advanced machine learning model to predict pricing trends and optimise sales. This model process 150 trillion combination possibilities to set the prices that maximise profit while ensuring customer satisfaction. Today, we explore 2 revolutionary applications in Fintech, algorithmic trading and fraud detection. Each of these technologies is reshaping how the finance sector operates, enhancing both efficiency and security. Let's start with algorithmic trading. Imagine 3D strategies that execute thousands of orders. A second strategies that can analyse and react to market data faster than any human. This is the reality of algorithmic trading, where AI algorithms use mathematical models to make lightning fast decisions, maximising profit and minimising risks. These algorithms take into account not only stock prices but also global economic indicators, company news, and even social media trends to make informed trading decisions. Moving to fraud detection. In a world where digital transactions are omnipresent, the risk of fraud is a major concern. AI steps in as a critical defence mechanism. By analysing patterns in a vast amount of transaction data, AI models can identify anomalies that signal fraudulent activities. These algorithms take into account not only stock prices but also global economic indicators, company news, and even social media trends to make informed trading decisions. Moving to fraud detection. In a world where digital transactions are omnipresent, the risk of fraud is a major concern. AI steps in as a critical defence mechanism. By analysing patterns in a vast amount of transaction data, AI models can identify anomalies that signal fraudulent activities.

Video 3: Types of AI

Artificial intelligence isn't just the single technology, but spectrum of capabilities and potentials. We'll explore three distinct types of AI: one, Artificial Narrow Intelligence or ANI, two, Artificial General Intelligence or AGI, and three, Artificial Super Intelligence, which is ASI. Each represents a different stage in AI development, each with its unique impact and potential applications. First, let's discuss Artificial Narrow Intelligence or ANI. This is the type of AI that surrounds us today. ANI specialised in tasks specific. Think of voice assistants, customer service, chat bots, and even recommendation systems on streaming services. Each is designed to perform a particular task and does it extremely well. A real world example is the use of ANI in healthcare for diagnostic assistance. AI systems analyse medical images to detect

diseases such as cancer more quickly and accurately than ever before. This not only speeds up diagnosis, but also increases the precision, enhancing patient outcomes. Next, we have Artificial General Intelligence or AGI. Unlike ANI, AGI can understand, learn, and apply knowledge across multiple fields. It's what we often see in sci-fi fields. An AI that can think like a human, solve general problems, and perform any intellectual tasks that a human being can. While fully functional AGI has not yet been achieved, advancements are being made.

An example in progress is autonomous robots being developed for unstructured environments like homes or small businesses, where they need to adapt to a wide range of tasks and scenarios. Lastly, we come to Artificial Super Intelligence, or ASI. ASI represents an AI that surpasses human intelligence and capability. It's an AI that can outperform the best human brains in practically every field, including scientific creativity, general wisdom, and social skills. ASI remains a theoretical concept, but one that drives discussions about the future of AI technology and its impact on society. The potential realisation of ASI prompts important ethical and safety discussions, ensuring that as we advance in AI capabilities, we also safeguard our societal values.

Video 4: Introduction to Machine Learning

All right. Let's plunge into the world of machine learning. Imagine you're teaching your dog a new trick, but instead of a dog, you have a computer in trick. Let's unravel this. Machine learning is fundamentally the science and yes, the art. Of course, programming computers to learn from data. Now, you might be thinking art, science, make up your mind, but that's the beauty of it. It's both. It's like teaching your computer to paint a masterpiece, but to paint this data and the masterpiece is, well, smarter decisions. Back in 1959, Arthur Samuel, a pioneer in the field, described it as the field of study that gives computers the ability to learn without being explicitly programmed. Imagine that it was like telling a computer here, read this data and figure something out and guess what it actually did. Now fast forward to 1997 and Tom Mitchell gave it an engineering spin. He said a computer program is said to learn from experience E concerning some task T and some performance measure P if its performance on T as measured by P improves with an experience E . In other words, it's like saying to a computer, you better get good at this or else. And the computer, like a diligent student, does just that. It learns, it adopts, it gets better. But why does this matter to us? Why should we care if a computer learns to play chess, recognise a face, or predict the stock market? Well, because machine learning is not just a futuristic concept. It's here. It's shaping our world, our decisions, our future. It's everywhere. From the recommendations you get on Netflix to the way Google Maps decides your route. So, let's dive deeper into this world of machine learning. Let's uncover the mysteries behind these self-learning machines. Are they just fancy

calculators or is there more than meets the eye? Let's find out. Together, let's embark on a journey from the old school to the new cool. Why machine learning you ask? Well, let's take a walk down the digital link to the era of traditional programming with our good old friend, the spam filter. Imagine in a not so distant past, crafting a spam filter was like being a digital detective. Words like free credit card, amazing and the notorious number four followed by you. Then with the patience of a St. you'd write algorithms crafting rules like an artisan and voila, your emails are spam 3 until the spammers evolve. Figure one. This is our traditional approach. It starts with studying the problem. Your knee deep in spam emails trying to find patterns. Next, armed with caffeine and determination, you write truths. If it says free, lock it. If it says amazing offer into the base it goes. But then the evaluation, it's a hit or a miss. And when it's a miss, it's back to the drawing board to analyse errors. It's a loop my friends, a task where the boulder is an endlessness of rules. Now cast your eyes on figure. This is where machine learning strides in. We will start by studying the problem, but instead of writing rules, we feed data into our ML algorithm. It's like giving a chef all the ingredients and saying make me a fist. And they do. They learn and whip up a recipe that adopts changes and improves the evaluation. Were often than not we are met with smiles, not frauds. And when there's a fraud, the ML algorithm tweaks the recipe, learning from its errors. Figures one and two show the transformation from manual toil to intelligent automation, from endless rules to elegant learning. This, dear learners, is why we embrace machine learning. Not just to predict the future, but to craft it with our own intelligent machines. Think of this as the rebirth of our spam filter. No longer. We are the rule writing scribes of your We've graduated to become the architects of the self improving systems. Here's how it works. We begin with the data, the DNA of our spam filter. It's raw, may be a bit messy, but ripe with potential. Enter Figure 3, the new era. We train our ML algorithm not just to read, but to understand and adapt. It's no longer about static lists, it's dynamic learning from patterns. Launch the model and it's out there. Learning in the wild for you becomes word for you. And our algorithm doesn't miss a bit. It adopts. It's the detective that's always on duty, tirelessly updating. It's know how. This is not just about automation, it's evolution in action, The spam filter of the past transformed into an ever evolving guardian of our inboxes. And that, my dear friends, is the power of ML. From rigid rules to a self updating shield against spam, a true digital Sentinel. Now let's see how this intelligence spreads across different domains, making not just our emails, but our entire digital world smarter and safer. Continuing from what we left our anti spam quest, Figure 4 illustrates the cyclical genius of machine learning. It's not just about the first Eureka, but an ongoing series of light bulb moments. Step one, study. We're not just looking at emails, we're understanding the spam's DNA. It's a detective work at its finest. Step 2, train. Here are ML algorithm fists on data, learning what's tasty, meaning legit emails and what's not, which is spam. Then we launch and observe. Is

our digital government out to par? We inspect like judges at the culinary show. And if it's not the perfect score, we define the beauty of ML. It's a loop, learning and improving with each pass. This, my friends, is the rhythm of machine learning. It's iterative, it's dynamic, it's relentless in outsmarting spam. Now let's move forward and see how this relentless rhythm lays out across industries.

Video 5: Types of ML Systems

Let's navigate the bustling marketplace of machine learning systems, where data is the currency and the algorithms are the traders, each with a unique expertise. Each system, each business in our ML marketplace specialises in converting data, our raw material, into actionable insights propelling industries and economies forward. As we prepare to dive deeper into this marketplace, consider how each of the systems can be leveraged to turn your data into your most valuable asset. Fresh from our marketplace tour, let's narrow down to specific vendors the types of ML by supervision. The four types of machine learning are reinforcement learning, supervised learning, semi supervised learning, and the last one, unsupervised learning. Reinforcement learning. Imagine a toddler tasting new foods. They experience joy and disgust, learning what they like and dislike through trial and error, eventually understanding what to choose for a reward. Supervised learning.

Picture a child learning to paint, guided by a parent. Each brass stroke is carefully instructed with labelled colours, creating planned masterpiece semi-supervised learning. Visualise a child learning with a coach at a distance. A coach provides occasional guidance, helping the child learn by doing with some structure and support, but also independence. Unsupervised learning. Think of a child exploring a world of blocks and ships without any instructions. They discover patterns and groupings through pure exploration and play. In the supervised learning quarter, we have the season stockbrokers algorithms like linear regression predicting future market trends based on the years of financial data. They are the Warren Buffet of the ML world, turning historical insights into future fortunes. Unsupervised learning. Meanwhile, it's adventurous explorer akin to a retail magnate, identifying new customer tribes, like discovering a top markets. Clustering algorithms segment customers by unseen patterns, not by labels provided in a survey.

Semi-supervised learning combines these worlds. Think of a consultancy firm that uses both comprehensive market studies and all the ground consumer interviews to provide the full picture of the business landscape. Enter the sports arena of RL. Here algorithms are like coaches defining game strategies after every match. They adopt plays and tactics based on the rewards of wins and feedback from losses. Next, our Learning Adaptability Logistics hub showcase online learning as the agile couriers make deliveries on the fly, where the model incrementally updates its understanding

as the new data arrives, allowing for real time adaptation. Batch learning on the other hand, represents the freight services planning extensive logistics for maximum efficiency.

In this approach, the model is trained on a large pre collected data set all at once, optimising performance before deployment in the financial districts skyscrapers. Instance based learning functions like analysts providing bespoke investment advice by comparing clients to past portfolios, relying on specific data points from past experiences to make decisions for new situations. Meanwhile, model based learning operates like forecasters who use complex algorithms to predict economic shifts, building a general model from data to make predictions about the future outcomes. In essence, these are not just types of ML.

They are approaches to learning as varied and rich as experiences that shape us from childhood. And just like us, they grow, they adopt, and they get smarter over time. Let's now delve into how these approaches play out in the sandbox of real world data.

Video 6: Supervised Learning

Picture yourself as the chief security officer in the digital headquarters of a global bank. Your task?

Safeguarding the email communication network against the cunning tricks of spam, ensuring only legitimate ham messages get through. Each email is an employee flashing their badge to the scanner. The labels are the security clearances, red for the threat, green for a go. Our supervised learning algorithm is the vigilant guard at the gate, trained to spot the difference between a genuine badge and a clever forgery. In this high stakes environment, attributes are the features of the badge, the colour, the shape, the hologram, feature is the specific detail, like a badge with a hologram that reads access level executive. It's the algorithm's job to check every badge against its training, to let the true message in and keep the deceptive ones out. Just as the security of the bank hinges on the keen eye of its officers, the integrity of our banks relies on the precision of our supervised learning algorithm. It's not just filtering emails, it's protecting the bank's digital vault. And so as we navigate through the cyber corridors of our bank, remember that supervised learning is the guardian that never sleeps, ensuring that every message we receive is there for the right reasons. And now from the Sorting Hat of Classification, we step into the fortune teller step of regression. Here, supervised learning doesn't just label, it predicts, like guessing the final score between the game has ended. On this canvas, we're not just looking at categories. No, we're painting a numerical masterpiece, predicting values like a car's price from the rich palette of its features, mileage, age, brand. Imagine each data point as a car on our lot. The feature one axis, that's the odometer reading, and the value we're predicting is the price tag hanging in the window. Our algorithm is the

savvy salesperson who, after seeing countless sales, knows just how to price them. But here's where it gets even more intriguing. Some algorithms, like the chameleon of logistic regression, can switch between predicting precise values to guessing the likelihood of a car being a vintage, a collectible, or a budget friendly. So, in this marketplace of machine learning, regression is your auctioneer setting the bill based on the whispers of features. It's a numerical soothsayer telling the value in the language of data. Armed with this foresight, let's cruise ahead to see how these predictions power decisions in the real world. From finance to your next vehicle purchase. Welcome to the corporate lead of supervised learning, where algorithms are the star players on the field of prediction and analysis. First up, the linear models. Think of a linear regression as the reliable accountant diligently tracking each dollar it's sent to predict your company's financial trajectory.

Logistic regression, or what we call LR on the other hand, is the sharp HR specialist, classifying applicants as hired or not hired with a keen eye on the trades that spell success. Then we have Ridge and Lasso regression, the financial auditors of the machine learning world. Ridge regression stabilises your stock portfolio against volatile market movements, while Lasso selectively zeros in on the most impactful factors, cutting through the noise to optimise your marketing spend. Transitioning to tree-based models, we have decision trees, the versatile project managers mapping out the pathways of decision making with a very clear cut choices and outcomes. Random forests amplify this, pulling the wisdom of multiple projects from teams to forecast market trends and customer behaviours. And then the elite strategist, the GBM or the gradient boosting machines, XGBoost and the LightGBM. These are the consultants with their sophisticated algorithms boosting your predictive power, whether it's forecasting sales or preempting maintenance issues with uncanny accuracy. These algorithms are the heavy hitters in your line up, each bringing their unique strengths to the plate, turning data into actionable strategies and clear business outcomes. With these top players on your roster, let's draft a winning strategy to take your business to the championship.

Video 7: Unsupervised Learning

Imagine you're a mastermind behind the bustling marketplace with customers, each with their own unique tastes and preferences. This is the essence of unsupervised learning, deciphering the silent stories of shoppers without any guideposts. Picture our marketplace.

Each shopper is a data point, a character in this bustling bazaar. Feature one might be their spending habits. Feature two could be their browsing time. There are no signs telling us who they are, no bargain hunter badges or luxury lover stickers. Our algorithm is like Shopperkeeper, observing from a distance. It sees patterns emerge

from clusters of customer form naturally here, a group that others around the sales rack. There's a click that lingers by new arrivals. It's a silent Symphony of segmentation. Without uttering a word. Our algorithm reveals a certain percentage of our visitors that are late-night luxury seekers, while another segment are weekend window shoppers. It's inside without intrusion, understanding without asking. And when we apply a sophisticated algorithm, our shopkeepers wisdom depends. He sees not just crowds, but the new ones, layers within subgroups with specific interests, all shopping under the same roof. This is isn't just a customer analysis. It's a regulation of hidden patterns that can transform how we stop our shelves, tailor our marketing, and personalise the shopping experience.

Unsupervised learning doesn't just give us data points, but it gives us a narrative as we close the doors on today's marketplace, we're equipped with knowledge to create not just products but experiences tailored from every subtle preferences of our clientele. Step into the control room of a financial powerhouse where transactions flow like a river. Among this anomaly detection anomaly detection is the vigilant Sentinel, the one that keeps the river clean from the pollution of fraud. On our display you will see the usual flow of data. These are typical everyday transactions in the lifeblood of the commerce. Feature one could be transaction amounts and feature number two might be represent the frequency of transactions per customer. You notice the cluster of normal. These are the transactions that follow predictable, comfortable patterns of routine business. But what's this?

An outlier, the red flag in our river of data. This is the crux of anomaly detection. Like a hawk eyed auditor, the algorithm spots which doesn't belong the potential fraud. It raises the alarm, allowing us to stoop in and investigate before the ink on a fraudulent check price. In a world where financial fraud can lurk in the shadows, anomaly detection shines a spotlight on the unusual, the suspicious, the potentially cost abbreviations in our financial narrative. As we live in the control room, let's carry with us the assurance that Anomaly Detection provides a watchful Gurchan ensuring the integrity of every transaction. Imagine you're the CEO of a tech savvy retail empire. In your toolkit, you have some of most astute advisors, each specialising in deciphering the secrets behind customer behaviours and multiple trends. These are your top unsupervised learning algorithms, the silent strategies of the retail world. First, the clustering with K-means leading the charge. It's like having a personal shopper for every customer, grouping them by their shopping patterns. Second is the HCA. The hierarchical cluster analysis goes even further, breaking down these groups into subgroups like separating casual shoppers from the deal hunters, expectation maximisation grounds at the team by not grouping but also finding the probability of membership. Essentially figuring out if a customer is probably a bargain hunter or likely a luxury seeker. Then we step into the art of gallery of visualisation and dimensionality reduction. Here, PCA is your creator, transforming a jumbled collection of art into a masterpiece of insight, reducing the complexity without losing the essence. Kernel PCA takes this into the next dimension, adding the layer into the

depth of the data, revealing underlying patterns not visible to the naked eye. LLE or local-linear embedding and Disney are they innovative artists, painting the picture of the data in a ways that highlight the most interesting and informative relationships.

And lastly, the association rule learning algorithms Apriori and Eclat, they are your market analysis, uncovering the hidden rules that drive sales. Customers who bought their swimsuit are often by a sunscreen. That's the kind of actionable insight that can reshape your cross selling strategies. These algorithms are your silent partners working behind the scenes, turning chaos into a data into the clarity of strategy, revealing the patterns that drive profits and customer satisfaction. As we close the Ledger on this overview, let's prepare to deploy these algorithms to unlock the potential within our data, driving growth and innovation.

Video 8: Semi-supervised and Reinforcement Learning

Envision a coffee shop where each patrons preference is a blend of subtle flavours. Now, meets semi-supervised learning. The barista who remembers not just names but tastes after the briefest introduction. Here's our coffee shop's ledger that's of data each a customer's order. A few marked with shapes, our regulars, the labelled data. The majority are anonymous patrons that are unlabelled, yet not unnoticed or barista. The algorithm starts with a few known favourites, then, through the magic of semi-supervised learning, begins to anticipate the orders or even the occasional visitors, turning a simple plus query into personalised coffee experience. This is the power of semi supervised learning. A touch of guidance leads to a wealth of understanding, transforming the sparse into the menu of predictions. Think of reinforcement learning as the world's most adaptive entrepreneur in the marketplace of challenges. Each choice, like selling the right investment, it leads to a triumph of or a teachable movement. Here's our agent in the thick of action, learning the hard way that not all investment are lucrative. Ouch, that's a market crash -50 points. But it's not about the set back, it's about the comeback. Much like an investor, our agent recalibrates after a loss. Tech stocks are volatile, and refines its strategy to maximise gains over time, turning market into lemons, into profitable lemonade.

Through a cycle of trials, the agent owns its policy, learning to dodge the flames of the risks and desist the buckets of opportunity, each iteration smarter and more resilient. The perfect formula for business group. This, ladies and gentlemen, is the essence of RL, or what we call reinforcement learning. Swift adoption, strategic decisions, and the relentless pursuit of success. AlphaGo's triumph over a gold champion was not just a victory of the machine over man. It was a showcase of an agent's journey to excellence. Learning, adopting and evolving strategies from playing countless games against itself. This is RL, or what we call reinforcement learning. A continuous loop of trial and error and triumph, where the ultimate reward is not just a high score, but a mastery over complexity.

Video 9: Challenges of ML – Part I

In the intricate dance of algorithms and datasets, machine learning faces its share of challenges. It's a path lined with pitfalls that can trip even the most carefully designed models. Let's navigate these hurdles together with a keen eye on the real world business implications. First, the quantity of training data sets the stage. Imagine a business eager to predict customer's behaviour but hindered by scant transaction histories. The lesson?

More data points can lead to more informed decisions. Next, we encounter a representation challenge. Consider a global app launch that only samples feedback from one region, skewing its understanding of user satisfaction. A diverse data set ensures that no voice is left unheard, no trend overlooked. Quality is king in the realm of data. A financial institution plagued with erroneous records risks faulty credit risk assessments. The antidote, a vigilant approach to data integrity, features the attribute chosen to train a model. It must be relevant picture retailer using a relevant social media metrics to forecast sales, only to find them disconnected from the actual consumer behaviour. Overfitting is like wearing blinkers, where a model sees the training data in a high resolution but misses the broader picture. It's like a stock analysis algorithm that's so attuned to past market fluctuations, it fails to adapt to an economic counter. Conversely, underfitting is akin to viewing the market through a foggy lens, where simplistic algorithm overlooks subtle market dynamics, rendering its predictions naive and often incorrect. Each of these challenges, if unaddressed, can steer a promising machine learning project, of course turning potential insights into misinformed strategies. But fear not, for with the right expertise and a rigorous approach, these obstacles become stepping stones to building models that not only learn but also adapt and thrive into the ever evolving landscape of business intelligence.

In the bustling world of machine learning, data is the new currency. The quantity does have a quality all its own. Imagine stepping in the world's most sophisticated car. But there's a catch. The gas tank is nearly empty. This is the predicament we face with insufficient training data in machine learning. Our journey today takes us through a landscape for algorithms like Memory-Based, Winnow, Perceptron and even Naïve Bayes thrive on the lifeblood of data. Look at this trajectory. As the volume of data increases, so too does our accuracy. It's a clear signal. The more fuel in our algorithms, the smoother, the further our right into predictive precision. Now let's talk business. In the real-world, company's ability to forecast trends, understand customer behaviour, or automate operations hinges on their very principle. Without sufficient data, even the most advanced algorithms have their hands tied their insights limited. But there's a plateau. This is where we must make a pivotal decision to invest in gathering more high-quality data or to innovate with what we have. The message is loud and clear in the fuel rays of machine learning, data is the key. And so I urge you,

let's not leave our machine's potential untapped. Let's fill the tap and prepare for a journey of transformation, innovation and unprecedented growth. Ladies and gentlemen, imagine you're planning the grand opening of a global Bistro with the menu that caters to taste buds worldwide. Now, you wouldn't just sample dishes from the nearest street corner, would you? No, because that would be like training a model in a world map where half of countries are missing. Here's a stark example analysing life satisfaction versus GDP per capita. If we only look at the subset of countries our model represented by this dotted line, we might tell us that more money always means more happiness. But when we invite the whole world to the party, adding in the missing countries, our new model, this solid line paints a picture a different picture. Suddenly, the relationship isn't so straightforward. It turns out our happiness recipe needs more than just a sprinkle of wealth. The richest countries aren't always the most satisfied, and some less wealthy nations are quite content. This is a classic case of sampling bias. It's like trying to predict the next foodie trend by only sampling fast food. Convenient, but hardly representative. So, as we gather our ingredients for a successful machine learning model, let's ensure our palm tree is stocked with a diverse array of data representing the full spectrum of experiences. Welcome to digital market, where data is the currency and the quality is the gold standard. Imagine you're launching a rocket. Your brand new app, your fuel. It's the data. But if that data is poor quality, it's like filling your rocket with the muddy water instead of refined fuel. You won't get to the stars that way. Like a GPS navigation app, one wrong turn and you're not exploring any crowds. You're exploring the wilderness. Cleaning data isn't glamorous. It's painstaking. It's behind these scenes of work of our digital world. Let's say you are collecting customer feedback. A mis-coded survey response is an outlier. It's not an exotic fruit in our data salad. It's a misplaced onion. And when there's missing information, like a blank H field, that's a gap in our rocket's insulation. Now do we patch it up or do we redesign the cabin? Remember, even in a world driven by algorithms, the human touch is invaluable. Scrubbing our data set is our commitment to craftsmanship. In the era of automation, let's graph our data with the precision of a master jeweller, ensuring that every piece contributes to the brilliance of the whole world. Now imagine you're a captain of the ship in the vast ocean of data, your compass feature engineering the journey behinds, with feature extraction, distilling row into a meaningful insights. The next we navigate through the waters of feature selection in here. We will make critical choices, what stays on the board and what's thrown overboard to keep our model swift and agile. Sometimes we even dive into depths of creativity to forge new features, crafting inputs that could be unveiled patterns previously hidden in murky waters of raw data. And as we funnel down these processes, we refine our requests, converging on the art of feature engineering.

Video 10: Challenges of ML – Part II

Now, let's anchor in the real world. Consider a financial firm assessing credit risks. It's not just about having data, it's about having the right data. Irrelevant features could be noise, like judging credit worthiness based on your favourite ice cream flavour. Does it make sense? Absurd, right? Through feature engineering, we shift out the chocolate chips from the vanilla noise, ensuring our model focuses on income, spending habits, and credit history, the real predictors of risks and just like that, we transform a sea of information into a stream of intelligence. So, let's embark on this journey with precision, casting away the irrelevant and steering towards the essence of predictive power. You see a graph that's trying too hard to follow the ups and downs of data points. It's like a financial model that's been tailored to predict stock prices based on past market fluctuations, including every minor spike and deep.

The problem here is that while the model might predict past prices with incredible accuracy, it's too sensitive to historical noise. These are the random events that affect the stock prices but are unlikely to repeat in the same way. When we use this model to predict future stock prices, we'll find it performing poorly because it's learned from the noise rather than the actual underlying trends. Now, let's say we're in a fintech company that's developed this model. Our clients rely on us for future market predictions, but they are quickly losing trust because our predictions are off. This is classic case of over fitting. Our model doesn't generalise. We're doing new unseen market conditions to address over fitting. We simplify our model. We remove the less significant variables that don't have a strong correlation with market performance. We also introduce regularisation techniques to penalise complexity. The result? A model that is robust and performs better on a real world data. It's not swayed by a random fluctuations of the pasts but rather captures the true patterns that drive market movements. Have you ever heard the saying too much of a good thing? Well in this world of machine learning this is known as over fitting. But fear not, today we are going to arm ourselves with three powerful strategies to keep our models learn mean and fit for the real world.

Let's dive in. First up, simplify the model. Imagine you're a chef and you're this is your model. Now, you would not want to overfire your signature dish with too many ingredients, would you? Similarly, we keep our model simple, focusing on essential features that truly impact our outcome. This features our model isn't overwhelmed by the culinary equivalent of unnecessary spices. Moving on, we have enriched the data set. Think of this as having more customers taste test your dish. The more feedback you get, the better you refine your recipe. Similarly, by gathering a larger and more varied set of data, our model learns from a richer set of experiences, becoming more adept of predicting the future outcomes. Lastly, we claim the noise. In our kitchen analogy, this would be like ensuring all your ingredients are fresh and of high quality

noise. Boiled tomatoes in our sauce. By cleaning our data, we remove misleading outliers and correct errors, ensuring our model isn't LED actually by relevant or incorrect information. In conclusion, by simplifying our models, enriching our data sets and cleaning our data, we can ensure our machine learning models perform not just well, but appropriately reflecting the true nuances of the real world data they are designed to interpret. It's about creating a model that's not just accurate, but also adaptable and reliable. Picture a sales forecasting tool that's designed to predict future revenue streams based on historical data. When the tool's predictive line is 2 straight, like our under fitted example here, it ignores the UPS and the downs of seasonal trends and market dynamics leading to inaccurate forecasts. This is the classic case of under feeding in the machine learning. It happens when our model is too simplistic to capture the complexities of the real world data. The straight line here makes sense to the untrained eye, but it misses out the nuances that could make or break a business strategy. To refine our model, we could incorporate polynomial regression. This allows for a curve that better fits our historical sales data, capturing those crucial peaks and values around holiday seasons or sales cycles. As demonstrated in the centre graph, a good fit acknowledges the variability in data over time, providing a realistic trajectory that business can rely on for strategic planning. By moving away from oversimplified models and embracing more nuanced approaches, we can deliver predictive analytics that truly resonate with the flows of dynamic market, paving the way for more informed decision making in the dynamic landscape of online retail. Imagine a company that uses a basic algorithm to forecast sales.

The model, however, is too simple and fails to capture the nuances of customer behaviour and seasonal trends, leading to inaccurate sales predictions and stock issues. Adopt A more advanced model to address this. We could shift from a linear to polynomial regression model or even machine learning approach like a decision tree that can handle nonlinear patterns, providing a more accurate fit for sales strengths. Improve feature selection by incorporating additional relevant data, such as marketing spend, website traffic, and customer reviews, we can give the algorithm a richer context for prediction, enhancing its accuracy. Relax model constraints. If our initial model was too restricted by regularisation, we just adjust these parameters. This allows the model to pay more attention to the training data, improving the ability to predict complex trends. By employing these strategies, the retail company can refine its forecasting model to better align with its business, leading to more accurate stock management cycles. As reduce wastage or shortages. Imagine launching a feature that suggests products to our users, increasing our sales and enhancing our user experience. But how do we trust the suggestions it makes? Our journey begins with test set usage. Like sending out a new ship into OpenSea, we assess our recommendation engine on data it has never seen before. We're looking for

performance not just in the calm harbour of our development environment, but in the unpredictable portion of real world data. The generalisation error is our compass here. It's not about how many products our model can suggest correctly in a controlled setting, but how accurate it can predict real customer preferences. How accuracy here means our customers find what they need without sifting through irrelevant suggestions. Next, we address the risk of avoiding overfitting a model fixated on the intricacies of retraining data, like a captain who can't adapt to changing ways it's done to fail. Through regularisation techniques, we trim the sails of our models, ensuring that it can navigate the complexities of real world using behaviour. Our quest continues with model selection and hyper parameter tuning. It's a process of selecting the right crew and equipment for our ship. We may need to adjust our sails, the hyper parameters to catch the right wings, ensuring the most efficient path to the accurate recommendations. Now, making informed decision requires a sound validation set. This step is like a trial run before the grand voyage. We tweak and define our model choices based on feedback, steering the clear of any racks of bias that might lurk beneath the surface and there comes cross validation, a technique as crucial as the rotation of duties on a long sea journey. It ensures every member of our data crew contributes, avoiding over reliance to any single pattern or trend. Finally, the final model evaluation. It is where we put our model to tests one last time. This is the moment of truth where we confirm if our recommendation engine can truly sail the high seas, bringing value to our customers and to our businesses. In the end, it's about setting sail with confidence, knowing our model is not just a but a robust ship ready to take into a vast data ocean.

Video 11: How AI Revolutionises Industries

AI and ML are more than just buzzwords, they are engine driving innovation across sectors, from healthcare to finance and from manufacturing to retail. Each industry has its story of transformation driven by the power of intelligent technology. Machine learning, deep neural network, pattern recognition, data mining, training on large data sets, basis for advancement in natural language processing or NLP. Computer vision and analytics. Natural language processing, translation, speech transcription, virtual assistants, sentiment analysis, language generation, content moderation, Computer vision, label extraction from image whether it's car, person, tree, etc. Face recognition. Image attributes, such as face gender, OCR to text, streaming video analysis, content moderation. Predictive analytics models that are usually embedded within reporting dashboards to enable real-time event detection. Support decision frameworks and conduct risk modelling and customer profiling. This provides insights to business leaders, helping them drive revenue, optimise operations, avoid loss, and improve customer service. User experience. The combination of text mining,

speech analysis, and NLP techniques can help improve customer engagement and user experience. One of the more common ways this is done is with the implementation of chatbots and conversational platforms. The simpler text chatbots do not perform advanced conversations or functions. Conversational platforms, on the other hand, can perform more complex functions and retain contextual information across a conversation. Advanced forms of interactions go beyond text to include speech, touch, gesture, and other human interfaces. Process automation: AI is being leveraged across a wide range of business process automation, from IoT project implementations of smart buildings and energy grids to automated contact centres, and supply chain and warehouse management using robots.

Understanding demand and having the capability to predict sales accurately is crucial for manufacturers, suppliers, and retailers. If the estimates are too high, the looming possibility of overproduction increases. While conservative estimates may lead to under production and lower customer satisfaction, in this case, achieving the right forecast at the right time could be a driver in attaining a sustainable competitive advantage for any company. The company has identified that sales volume forecasting is a cornerstone in running the operations of the business. Forecast generated by the sales team dictate many business activities such as raw materials acquisition, logistics, and even business expansion. Currently, the company utilises a three-month moving average to forecast the sales for a given week. The accepted forecasting methodology of the company has led to occasional underproduction or overproduction of various products sold. Given the situation, it is shared interest between stakeholders and data science and AI team to craft machine learning and deep learning models that can perform better than the current method. This enable the company to perform better decision-making in terms of operations with a whooping 2.5 million USD annual cost savings. This shows that the process done to create different machine learning and deep learning models for various products sold by the company. The first three steps of data cleaning and filtering were done. Some activities done in data cleaning were the mapping of sales region, renaming of products, and imputing null values of sales regions. After the cleaning step, a filtering process was done to create a product-region pair based on the top product and top product-regions per product using product-region pairing. Aggregation was done on a daily transactions to develop a weekly sales per product region pair. The weekly sales are preprocessed using Fourier transform and fed into various machine learning and deep learning models.

Now, data-driven forecasting models anticipate changing demand from different factors such as: one, frequency threshold; two, sales window; three, weather; four, holidays or no holidays or even shutdowns. As a result, we've lowered the operational cost by improving the accuracy of forecast related to demand and price. Without deep learning, it will take a lot of time to simulate 158 trillion combination. The cutting-edge AI deep learning architectures that outperform traditional time series forecasting methodologies. With this, we're able to come up with 97% accuracy in 4-

8 weeks implementation and iterations. To reduce the number of combinations we have to go through, we decided to use Genetic Algorithm or GA. GA is based on the theory of natural evolution by Charles Darwin, wherein the fittest individuals are chosen for reproduction in order to produce offspring of the next generation. Market and customer segmentation involves dividing a broad customer or business market into subgroups of customers based on some type of shared characteristics. These segments can be defined by criteria like demographics, behaviour, or even specific needs. Let's consider a real-world application with a major online retailer. This company, facing stiff competition and diverse customer base, turned to AI to redefine their segmentation strategy. Using AI and machine learning, they analyse vast amounts of data, from purchase histories to social media activity, to identify unique customer segments. The AI models deployed by this retailer could predict customer preferences and purchasing patterns with remarkable accuracy. They segment their market not just by traditional demographics but by nuance behaviors and preferences like price sensitivity, brand loyalty, and even preferred shopping times. By implementing these advanced segmentation techniques, the retailer was able to customise their marketing messages, tailor their promotions, and adjust inventory to meet specific demand. The result? A 30% increase in customer engagement and a 20% rise in sales within the first six months. In an era where customer interaction is increasingly digital, businesses face a dual challenge. Improving user experience while enhancing security measures, automated customer authentication addresses both by using advanced algorithms to verify identities quickly and accurately, reducing friction and increasing security. Consider a leading global bank that implemented biometric authentication across its digital platforms. By integrating technologies such as facial recognition and fingerprint scanning, the bank can now authenticate transactions and logins without manual intervention. The process begins with the customer attempts to access their account or initiate the transaction. The system uses a pre stored biometric data as a benchmark to compare real-time inputs. If the data matches, access is granted instantly. This shift not only streamlined the authentication process but also significantly lower the risk of fraud. Within the first year of implementation, the bank reported a 40% decrease in account takeover incidents and a 25% improvement in customer satisfaction scores related to account access. Now imagine applying similar technology in your industries. Whether it's retail, healthcare, or any service that requires a technicality verification, the potential to enhance both security and customer experience is fast.

Video 12: Ethical Challenges of AI

Artificial intelligence has become increasingly integral to various sectors, from healthcare to finance, promising transformative advances. However, as systems become more embedded in our daily lives, several ethical concerns have emerged that must be addressed. These concerns include gender bias, lack of transparency, discrimination, and privacy issues.

Gender bias AI algorithms can perpetuate biases present in training data, leading to discriminatory outcomes. For instance, search engines may deliver bias results based on gender stereotypes the enforcing existing prejudices online. Lack of transparency. AI decisions are not always intelligible to humans. Lack of transparency makes it challenging to understand how AI arrives to a specific conclusion or recommendations. Ensuring transparency in AI tools is essential for accountability and trust. AI based decisions can lead to discriminatory outcomes due to bias, data sources or embedded bias. Fairness and equal treatment should be prioritized in AI development and deployment. Lastly, privacy and security concerns. The security and privacy of data are critical ethical issues as AI rapidly develops. Safe guarding user privacy and ensuring data security are essential for responsible AI use. AI represents significant ethical challenges, including the perpetuation of gender bias, lack of transparency in decision making, risk of discrimination, and privacy concerns. Addressing these issues is essential for ensuring that AI systems are fair, accountable, and secure in their applications.

Video 13: Module Summary

We have reached the end of this module, where we explored the transformative power of artificial intelligence, what we call AI, and machine learning, or what we call ML, diving into their foundational concepts, real-world applications, and ethical considerations surrounding these technologies. We started by breaking down the basics of AI and machine learning. You've learned that AI refers to machine designed to mimic human intelligence, while ML is a subset of AI that allows systems to learn and improve from experience without explicitly programming. We also examined various types of AI systems and how they are already reshaping industries such as healthcare, finance, and retail. The module introduce you to the world of data science, which forms the backbone of AI and ML. You saw how data science combines business knowledge, computer science, and statistics to extract actionable insights from vast data sets. By leveraging this discipline, businesses can create smarter strategies driving innovation and operational improvements. We then cover the different types of machine learning, including supervised, unsupervised, semi-supervised, and reinforcement learning. In supervised learning, you saw how algorithms are trained using labelled data, while unsupervised learning enabled algorithms to detect patterns in unlabelled data. Semi-supervised learning offered a blend of both, and RL, as demonstrated through AlphaGo, allowed systems to learn through trial and error, defining decisions over time. Throughout the videos, you witnessed several real-world applications of ML, such as how Wiz uses real-time AI to predict traffic. Netflix recommends personalized content and Facebook identifies faces using advanced face recognition technology. These examples highlighted the practical integration of AI and ML in everyday life, enhancing user experiences and driving business efficiencies.

We also discussed the challenges associated with AI and ML, such as overfitting, underfitting, and data quality issues. You saw how insufficient or bias data can lead to poor model performance, while proper data management and model tuning are essential for achieving accurate predictions. Lastly, we address some critical ethical concerns surrounding AI, including gender bias, discrimination, privacy issues, and the lack of transparency in AI decision-making processes. You've learned why it is crucial to mitigate these risks to ensure that AI systems are fair, accountable, and secure. This module has given you a comprehensive introduction to the key concepts, applications, and challenges of AI and ML. You are now equipped with a foundational understanding of how these technologies are shaping industries and the importance of addressing their ethical implications as they continue to evolve.

Video 14.1: Demo: Predicting Real Estate Trends (Part I)

Good day. So, I'm here to present to you about the simulations regarding the cracking the code, it's more on predicting the real estate trends through AI simulation. Let's dive deeper. So, as you can see right now, you have the dashboard delinquency prediction. And normally, you have a new data. With regard to the new data, just like the one that you're seeing right now, you have different parameters. And for us to be able to determine the prediction when it comes to delinquency, we are going to click on 'Predict mode'. And from this, you can see that each and every name or parameter is being predicted depending on whether it's delinquent or not. So, how do we do that?

This is what we're going to discuss for today in terms of our simulation. Let's dive deeper. So, as an overview, this is a Real Estate Consumer Delinquency Prediction and why this specific simulation matters because in real estate, we're talking about financial stability and timely payments for us to be able to predict whether a certain customer is delinquent or not. One of the focus of this specific simulation is we were able to identify and mitigate the risks for us to be able to also optimise the resources and essentially enhance the customer retention. So, what you're going to learn in this specific session is we'll do data exploration and analysis. You will understand the specific of the dataset. You'll understand all of the details that pertains to the demographics, the financial attributes and even the impact on the frequency. Second is you will focus, we will focus on processing and feature engineering, which is how do we now prepare the data in terms of machine learning, preprocessing, the techniques like scaling, encoding and any other techniques such as handling imbalances or imbalance.

Third, is we're going to discuss the model training evaluation for us to be able to compare from the common model or the the the basic model going through complex model. And from that we need to be able to fine-tune for optimal performances or parameters. Then lastly, will generate insights and prediction based on what you have seen a while ago, we are going to operationalize the models that we are developing or we're going to develop. And from this specific simulation, we will translate that into

raw predictions and create meaningful business strategies. Again, it will be a real context. And what I prepared for you guys is this is really a pragmatic use case. In fact, I generated the data coming from the anonymous data that we have in, in a real business context. Now, let's drill down deeper. As you can see, right now we're talking about the groundwork for specific type of simulation. We will be using different libraries, Python libraries and AI and ML libraries. And later you will see also custom packages, custom packages. First and foremost, we are going to identify and develop and integrate our packages as you may have see, I'm using my own installed packages such as the CLAIM and modelling simulation version that 0.1. Eventually you will be taught when it comes to how do you now compare and organise in such a way that you have different codes and different libraries and organize it in such a way that you have your own packages or your own package. But for simplicity, I'm going to discuss first that this specific simulation will use a custom package that I did and share with you. Then we are going to import the libraries and packages.

As you as you can see right now, we have standard libraries. We have numerical and data libraries. We have visualization libraries, such as interactive libraries, Plotlib and other Seaborne libraries. You have also the machine learning model imports. The one that you are seeing right now are the main or common ML models. Then we have also pre-processing and feature engineering. From this feature engineering, we will be able to discuss different types of training, split and different scaling techniques as well as one hat encoding techniques. And from there we were able to discuss and create pipelines to be an input of our model. Then you will have model evaluation. There are different parameters that we are going to discuss in this specific simulation from the classification report to confusion matrix to accuracy score and so on and so forth. Then as part of the output of this specific simulation, we are going to use a Dash application or what we call Dash app for us to be able to integrate our application in the real-world context. And this specific application is being used just like the one that you're seeing right now. It, it will be used by the business or by your stakeholder or by your users for them to be able to again, update their specific dataset and, and predict that specific data that you have. And eventually it's really dynamic. Now, let's first discuss this specific library or custom library. Again, going back to what I mentioned a while ago, we will be using a custom library that I built for you guys for this specific simulation. The library that we're going to use is what we call `AI/ML_libraries.py`. As you can see, there are different libraries, standard libraries, numerical libraries and some of the other libraries that I mentioned a while ago.

But the core of this specific libraries are different classes. Different classes that we are going to use for each method or for each specific step or piece of AI and machine learning model, such as the EDA library or exploratory data analysis. Second will be what we call modelling libraries. Third will be MLPipeline libraries. The fourth one is what we call our this one evaluation libraries and lastly, our delinquency prediction dashboard library. Now, let's first identify what are the attributes of our data. So, these are real estate custom data prediction dataset. As you can see, the data dictionary that

is currently being used for this specific data is you have fielding and description. Based on this specific fielding, you can see that there are project phase that includes your phase one, phase two and phase three. Purpose, whether a specific contract is active or inactive. The the year of sale, whether what specific year the sale transaction was made, the month of sale from January, which is equivalent to year one going through December, which is equivalent to 12. The contract signing, the date where the contract was signed, the lot number, this one is a unique identifier for the lot being purchased or what we normally have in residential or lots. But when it comes to condominiums, it's condominium number. Next is household model, the type of household model purchase, whether it's a studio, whether it's one bedroom, two bedroom and penthouse. Space contract, the total amount stated in the contract. In this specific real pragmatic dataset, it's in PHP, as you may have seen in my previous slide, it's in peso, Philippine peso. Then the payment scheme, this is the payment method chosen by the customer whether it's spot cash or bank financing. Then the DP percent is it's, what they call 1 down payment paid by the customer. Then the months for take out, the number of months until a property take-out or transfer. That's the normal process with regard to real estate process or basis. Then list price, that is the list price of the property.

Again, in Philippine peso, the paid in, in principal or paid on principal, the amount paid on the principal. The nationality again, it's the nationality of the customer, whether it's a Filipino, American, Korean and so on and so forth. The marital status states the marital status of the customer, whether it's single, married and what not. The age, the age of the customer in years, the gender, the gender of the customer, whether it's female or male. The city where the customer resides, normally it's sitting here in the Philippines such as Makati, Taguig, etc. The location tag, the tag indicating the location category, whether it's prime or emerging market. The lot type, which is the type of the lot purchase, whether it's residential or commercial. And lastly, the count tag, again this is our target variable whether it's on not delinquent or delinquent.

Video 15: Demo: Predicting Real Estate Trends (Part II)

Now, let's go to the next step that is your data processing. When it comes to data processing, we will identify what is the data all about and from this specific type of simulation, we are going to load our data. Now, let's run that specific data. As you can see, there are 22 columns from project phase. Going through your project page on principle, again, you will learn those type of commands in your Python module. But for simplicity, what we are going to discuss right now is we have a real estate data in CSV format, and from that CSV format, we are going to translate that into our data frame or what we call pandas data frame. From that pandas data frame, you will see that we are checking the top, the top five when it comes to the entries of that specific real estate data set, as you may have seen in this specific preview. Then we are going

to identify the information that is related to our data frame or what we call data. Both are your columns, as mentioned, the null counts and the object type, whether it's an object, it's an integer, it's float etc. Then, from that, we are going to determine whether there are missing variables, or even non-variables, or null variables. So, from this specific code, you will see that it specifies the check missing values. Again, going back to what I mentioned a while ago, we will use the libraries, the real estate EDA libraries. The first one that we used is the load data. The second one is the check missing values. Again, it specifies and checks the missing values of a certain data set, whether it's a null, what are the columns that are missing, etc. From this specific simulation, you will see that there are no missing values or again our data set is complete and probably you may have seen based on our data that info that there are no null counts. Hence our data is good for our analysis. Now, let's do exploratory data analysis. So, why do we do that? Again, the first thing that we identify as the benefit of the EDA is to understand the data set, to identify the relationships of those specific features or data set and highlight if there are issues. One good example in our case is number one, what is the distribution of the numerical features like our list price or down payment percentage or how do we, how do categorical variables like city or payment scheme vary across data, across data set?

And lastly, are there any correlations between features that we would we should consider? So, let's first tackle the categorical distribution. From this specific categorical distributions, we will use visualise categorical distribution. Why? Because we are going to the determine if there are anomalies as regards to our distribution of the city. In this specific distribution, we call the the plot leaf for us to be able to determine and plot in bar chart the different cities. The frequency of each city are the one that you're seeing right now in this specific notebook. Then, from there, we are going to check also the numerical distributions. From the numerical distributions, what we did is going to the actual implementation. We are going to visualise the distribution of our numerical column. Again, we are going to determine whether there are historical relationships or distribution of each column or does that relate to your list price. From this specific slide, you can see that from the two million list price going through the roughly 15.25 million, that's the normal, normal range of our list price and these are the distributions. As regards to the median, our median is 8.45 million. In terms of our minimum, we have two million and when it comes to our maximum, we are talking about 15, almost 15 million these spreads. Then let's go now to our correlation analysis. When we do correlation, we are more focused on numerical data types. So, let's use our analyse correlation. What does it mean? We're going to visualise the correlation matrix from specified numerical columns and from this specific method. I used correlation matrix as an example and from that specific correlation matrix I'm visualising whether what's the relationship of each variable just like the one that you're seeing right now from the year scale going through your pay on principal values and from this specific parameter, you can see that like for this specific instance this there is a correlation of 0.78 as regards to sales contract versus the paid-on-principal. So, what does it mean? It means that there is a strong

correlation of the paid-on-principal versus your sales contract. Another one is as regards to your sales contract and list price, there is a +1.0 correlation, which is very strong correlation, almost perfect. Again it's perfect correlation. Then, from that we are now identify and visualise the relationship to identify potential clusters or separations in the data. So, let's do that. When we do pair wise, we are using pair plot. How do we do that? From pair plots, we identify using our seaborn to identify each plot just like the one that you are seeing right now. As you can see, the correlation that we identified a while ago is similar to this one. Like sales contract, you have positive linear relationship as regards to your list price. So, then again you can see different variations, different relationship for each parameter or feature. Now, since we have a very good relationship and we were able to identify the relationship and in variable, now we are ready, ready for pre-processing of the machine learning. So, how do we do that? We prepare the data set for our AI and ML models. We are going to clean and structured data ready for machine learning. We are going to determine whether a data is balanced and if it's not, we are we will use different strategy for us to make it balanced. The lastly, we will compare and understand the baseline accuracy. So, let's do first a calculating the relationship or the proportional chance contract. Another or proportional-by-chance accuracy. So, how do we do that? Let's do our ML pipeline class. From that ML pipeline class I created, calculate the relationship proportional-by-chance accuracy. From that specific calculation you can see that each plus will be calculated in terms of proportional chance. So, it's like our what is the chance or probability that it will happen? And if we go back to each particular parameter you will see the population per class. So, what does it mean if you have a plus that is not delinquent? When it comes to specific class, we have 679 class, for delinquent, you have 329 count and when it comes to proportional chance accuracy rate, we're talking about 57 percent or not delinquent and almost 12 or 13% as regards to delinquent and this is the one that you're seeing right now specific to your specific to your graph, the relationship proportional-by-chance accuracy rates and how do we do that? If we combine all of those proportional chance criterion, we are talking about 70%. So, what does it mean? You have 70% as your B sign accuracy. So, your model should be able to beat that baseline accuracy. It should be higher than your 70% proportional chance criterion and this is emphasising this specific item. Now, let's handle again missing values. In checking again we will fill in if there are missing values then we will use encoding for categorical variables. So, what do we use for preprocessing? So, we, I use the pipeline to be able to identify what is the correct preprocessing. For categorical variables I use one hot encoding and again you will learn those strategy in your succeed succeeding modules. And for numerical variables or features I used min-max. Min-max because later you will see that I'm after about specific distribution for this specific simulation. Then you will see that I was able to encode categorical variables using one-hot encoding. Then from that I was also be able to identify and use and scale the variables again. What is the beauty and what is the value of that one? For this specific numerical valuables, it retains interpretability using min-max and it aligns with non-normally distributed data is for real estate that

is non, non-normally distributed data. Then as you can see from this specific slide a while ago, our data is imbalanced because you have not delinquent of 679 and you have delinquent of 321. So, what do I do? What I do is I make and handling the class imbalance and use balance method. How do I do that? For the balance method I use SMOTE. SMOTE is another methodology or method that you will learn moving forward for you to be able to balance a data set. So, using a SMOTE or Synthetic Minority Oversampling technique you will generate syntactic examples of the minority of the class. Parents, right now you can see that the delinquent is 679, and the non-delinquent is also 679. So, it's already balanced. Then, from that you we will do splitting. Splitting is splitting your data set from test and train data set. In our case, we will split that using our test size 0.2. So, from this one, the test size or the training size will be 80% and your test size will be 20%. In our case, our test site, or training test size is 1,086 and now our testing set size is 272.

Now let's do model training and comparison. In this specific models, I created ML models such as logistical regression, decision tree, random forest, and what we call SVM or Support vector machines. I also integrated decision tree such as again random forest and gradient boosting and XG boost. And lastly, a more complicated model such as LightGBM or Light Gradient boosting method. So, from that, I was able to compare different models using different parameters. As as you can see, we are comparing a model that related and the results of the training accuracy and again the test accuracy. From that you will, you can see that the most common ones, or I mean the good ones are the random forest and the support vectors machine. So, what I'll do is I'll do hyper tuning parameters. Hyper tuning is, it generates well for us to be able to avoid over fitting and under fitting and it also improves the performance of our model. From that, you will see that it's now running hyper parameter from those models that I was mentioning a while ago. Then, right now, as you can see, it's using let's let's go to our shown best parameter. It's using different parameters and hyper parameters based on decision tree, random forest, gradient boosting, etc, for us to be able to select what is the best model and from that specific more than evaluate the corresponding test parameter.

Now it's running, and going back, it will select the best model and the parameters that is related to that specific model. Again, it's using Grid Search CV for the best model parameter. Now it's running and it's almost there, actually. Again, these are the methods that you will learn eventually in your respective modules. As you can see, the resulting model is 0.81 up 81.25%. Hence, going back to our baseline accuracy, comparing it to our 70% accuracy, our model is good and well-versed compared to our previous model. And this is the one that you are seeing right now. Based on our model, as you can see from the recall, from the precision F1 score, you can see that our model is better compared to the proportional chance criterion. In fact, from the delinquent, which is 0.58, our model is performing from 90 to almost 77% and from 0.13%, we're talking about 0.87 to 79%, just like the one that you're seeing right now. It's performing well when it comes to delinquent and when it comes to class one, it's also



performing well. And comparing it to our proportional chance criterion, it's, it's really good when it comes to precision and recall and it far exceeds the baseline. And due to time limitation, I'm now simulating the one that you saw a while ago. In our case, we are now using that and operationalising it so that eventually our user will be able to determine and predict those parameters and models. Again, the models, the preprocessor, just like the one that you're seeing right now, are saved into PQL for us to be able to operationalise and use those models. With that, that is our simulation for real estate delinquency prediction.